

TEA LEAVES DISEASE DETECTION USING DEEP LEARNING

by

ADHITHYA VARDHAN

21BAI1535

N. ANIRUDDHAN

21BRS1682

D.VISHAAL

21BRS1173

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VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Vandalur – Kelambakkam Road

Chennai – 600127

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ABSTRACT:

Tea cultivation faces significant challenges due to various diseases, including gray blight, blister blight, and red rust, all of which significantly impact both the yield and quality of the tea produced. Traditional detection methods for these diseases are often labor-intensive and subject to human error, making them inefficient and subjective. This paper delves into the application of deep learning algorithms, such as VGG16, MySequential, and Inception, to create automated tea leaf disease detection systems. Convolutional Neural Networks (CNNs), particularly VGG16, offer robust feature extraction capabilities that help detect diseases with high accuracy. MySequential and Inception architectures further enhance model performance through deeper network layers and a more efficient utilization of computational resources. Transfer learning is a key technique used in this approach, allowing the adaptation of pre-trained models to tea leaf disease datasets, thereby addressing challenges posed by data scarcity. Additionally, augmentation techniques and advanced neural network architectures contribute to precise disease localization and segmentation within tea leaf images. The integration of deep learning algorithms into tea leaf disease detection systems promises to deliver timely insights for farmers, enabling them to adopt proactive disease management strategies and promote sustainable tea cultivation practices. Ongoing research in this field remains essential to overcome existing challenges and fully realize the potential of deep learning in revolutionizing tea leaf disease detection and enhancing agricultural resilience. Through continued innovation, deep learning can transform traditional agricultural practices, making them more efficient, accurate, and sustainable.

INTRODUCTION:

Tea cultivation faces a multitude of challenges, with plant diseases such as gray blight, blister blight, and red rust posing significant threats to yield and quality. These diseases not only reduce the quantity of tea leaves but also impact their quality, leading to economic losses for tea farmers. Traditional detection methods, which often rely on manual inspection, are time-consuming and subjective, with varying levels of accuracy depending on the inspector's experience. Given these limitations, there is a growing interest in using deep learning as a transformative

solution, offering automated detection of tea leaf diseases from digital images with greater accuracy and efficiency. This paper delves into the potential synergy between deep learning algorithms and tea leaf disease detection, exploring how technology can transform traditional agricultural practices.

Deep learning, particularly through Convolutional Neural Networks (CNNs), has proven to be highly effective in extracting intricate features from images, making it well-suited for detecting diseases in tea leaves. CNN-based architectures like VGG16, MySequential, and Inception have demonstrated considerable success in classifying tea leaf diseases with high accuracy. These architectures consist of multiple layers of convolutional filters that can identify patterns associated with specific diseases, enabling them to distinguish between different types of leaf damage. Transfer learning plays a crucial role in this context, allowing the use of pre-trained CNN models, such as VGG16 and Inception, and fine-tuning them with smaller tea leaf datasets to address data scarcity challenges. Moreover, advanced data augmentation techniques and specialized architectures facilitate more accurate disease localization and segmentation, leading to precise identification and targeted interventions.

The integration of deep learning into tea leaf disease detection systems promises to provide farmers with automated, timely insights, enabling them to take proactive measures to manage diseases. This shift towards automation can significantly reduce the workload on human inspectors while offering greater consistency and reliability. However, this approach also presents challenges, such as the need for larger datasets and issues related to model interpretability. Ongoing research and innovation are critical to addressing these challenges and fully unlocking the potential of deep learning to revolutionize tea cultivation practices. With continued advancements in this field, deep learning has the capacity to drive sustainable tea production, enhance agricultural resilience, and contribute to more robust food security.

LITERATURE REVIEW:

S.no	Title	Year	Author	Methodology
1	TEA LEAF DISEASE DETECTION	2023	Nishmitha, Nikitha Shetty	Overall, the outcomes of research on tea leaf disease detection are encouraging and suggest that automated methods using image processing and machine learning techniques can be efficient in correctly identifying and classifying tea leaf diseases, potentially assisting in increasing crop yield and minimising financial losses for tea growers. According to this study, the Convolutional Neural

				Network classified diseases with a 92% accuracy rate. And support vector machine achieve 88.9%.
2	Detection and identification of tea leaf diseases based on AX-RetinaNet	2022	Wenxia Bao, Tao Fan, Gensheng Hu	Experimental results show the detection and identification accuracy of AX-RetinaNet for tea leaf diseases in natural scene images is better than the existing target detection and identification networks, such as SSD, RetinaNet, YOLO-v3, YOLO-v4, Centernet, M2det, and EfficientNet.

				The AX-RetinaNet detection and identification results indicated the mAP value of 93.83% and the F1-score value of 0.954. Compared with the original network, the mAP value, recall value, and identification accuracy increased by nearly 4%, by 4%, and by nearly 1.5%.
3	<i>A Novel Approach For the Detection of Tea Leaf Disease Using Deep Neural Network</i>	2023	<i>Saikat Datta, Nitin Gupta</i>	<i>The suggested method demonstrates that the model can accurately detect the kind of persistent tea leaf disease with a 96.56% accuracy. The accuracy of</i>

				the following disease classes are as follows, Algal Spot has an accuracy of 98.23%, Brown Blight has an accuracy of 97.98%, Gray Blight has an accuracy of 93.46%, Healthy classes of leaves has an accuracy of 99.10%, the Helopeltis disease class has an accuracy of 98.98% and Red Spot has an accuracy of 92% The model that is proposed in this literature is far superior than the existing methods in terms of accuracy.
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4	<i>Tea leaf disease detection and identification based on YOLOv7 (YOLO-T)</i>	2023	<i>Md. Janibul Alam Soeb, Md. Fahad Jubayer, Tahmina Akanjee Tarin</i>	<i>This study incorporates data augmentation approaches to solve the issue of insufficient sample sizes. The detection and identification results for the YOLOv7 approach are validated by prominent statistical metrics like detection accuracy, precision, recall, mAP value, and F1-score, which resulted in 97.3%, 96.7%, 96.4%, 98.2%, and 0.965, respectively. Experimental results demonstrate that YOLOv7 for tea leaf diseases in</i>
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				<i>natural scene images is superior to existing target detection and identification networks, including CNN, Deep CNN, DNN, AX-Retina Net, improved DCNN, YOLOv5, and Multi-objective image segmentation</i>
5	<i>Image Analysis and Detection of Tea Leaf Disease using Deep Learning</i>	2020	S. Gayathri; D.C.Joy Winnie Wise	<i>Harnessing the formidable power of Convolutional Neural Networks (CNNs), specifically inspired by the pioneering LeNet architecture, we've successfully cultivated a paradigm-shifting model for Image</i>

				<i>Analysis and Detection of Tea Leaf Disease. Our meticulously crafted neural network, influenced by LeNet's foundational principles, has achieved an astounding 92.7% accuracy.</i>
6	<i>Identification of tea leaf diseases by using an improved deep convolutional neural network</i>	2019	<i>Hu Gensheng , Yang Xiaowei , Zhang Yan</i>	Accurate and rapid identification of tea leaf diseases is beneficial to their prevention and control. This study proposes a method based on an improved deep <u>convolutional neural network</u> (CNN) for tea leaf disease

				identification. Experimental results show that the average identification accuracy of the proposed method is 92.5%, which is higher than that of traditional <u>machine learning</u> methods and classical <u>deep learning</u> methods.
7	Automatic Recognition of Tea Diseases Based on Deep Learning	2020	Jing Chen Junying Jia	In this paper, a deep convolutional neural network named LeafNet capable of recognizing the seven types of diseases from tea leaf disease images was established,

				with an accuracy of up to 90.23%, aiming to provide timely and accurate diagnostic services in the remote and topographic tea plantation in China. At the same time, the traditional machine learning algorithm is applied for comparative analysis, which extracts the dense scale-invariant feature transform (DSIFT) of the image and constructs the bag of visual word (BOVW) model to express the
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				image based on the DSIFT descriptor. The support vector machines (SVMs) and multilayer perceptron (MLP) were used to identify tea leaf diseases, with an accuracy of 60.91 and 70.94%, respectively.
8	Detection of Disease in Tea Leaves Using Convolution Neural Network	2020	<u>Shyamtanu Bhowmik</u> <u>Anjan Kumar Talukdar</u>	The proposed works aim to detect tealeaf diseases using the easy method while retaining minimal computational complexity and resources, resulting in fast and accurate results

				because the convolutional neural network automatically extracts features for classifying input images [54,55]. Approximately 95.93% of precision is achieved with the developed model in experimental results .
9	Machine Learning-Based Tea Leaf Disease Detection	2023	Faruk Ahmed, Taimur Ahad	The first group of researchers used the conventional Vision Transformer (ViT) to detect tea leaf diseases automatically at the field condition, which mitigates the economic loss. They employed

				Vision Transformer (ViT) because of its superior accuracy, adaptability to multispectral data, and scalability. ViT was excellent in capturing complex image patterns, making them invaluable for precise plant disease diagnosis. Due to its versatility and efficiency, authors preferred it over conventional CNNs in plant disease categorization. Among these researchers, Boukabouya et al. (2022) had the highest accuracy of
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				99.7%, whereas Silva & Brown (2023) had the lowest accuracy of 90.02%.
10	Grey Blight Disease Detection on Tea Leaves Using Improved Deep Convolutional Neural Network	2022	J. Arun Pandian, Sam Nirmala Nisha	Image transformation techniques such as principal component analysis (PCA) color, random rotations, random shifts, random flips, resizing, and rescaling were used to generate augmented images of tea leaves. The augmentation techniques enhanced the dataset size from 1320 images to 5280 images. The proposed DCNN model was trained

				and validated on 5016 images of healthy, grey blight infected, and other diseased tea leaves. Classification accuracy, precision, recall, F measure, and misclassification rates of the proposed DCNN are 98.99%, 98.51%, 98.48%, 98.49%, and 1.01%, respectively, on test data. The test results show that the proposed DCNN model performed superior to the existing techniques for tea grey blight disease detection.
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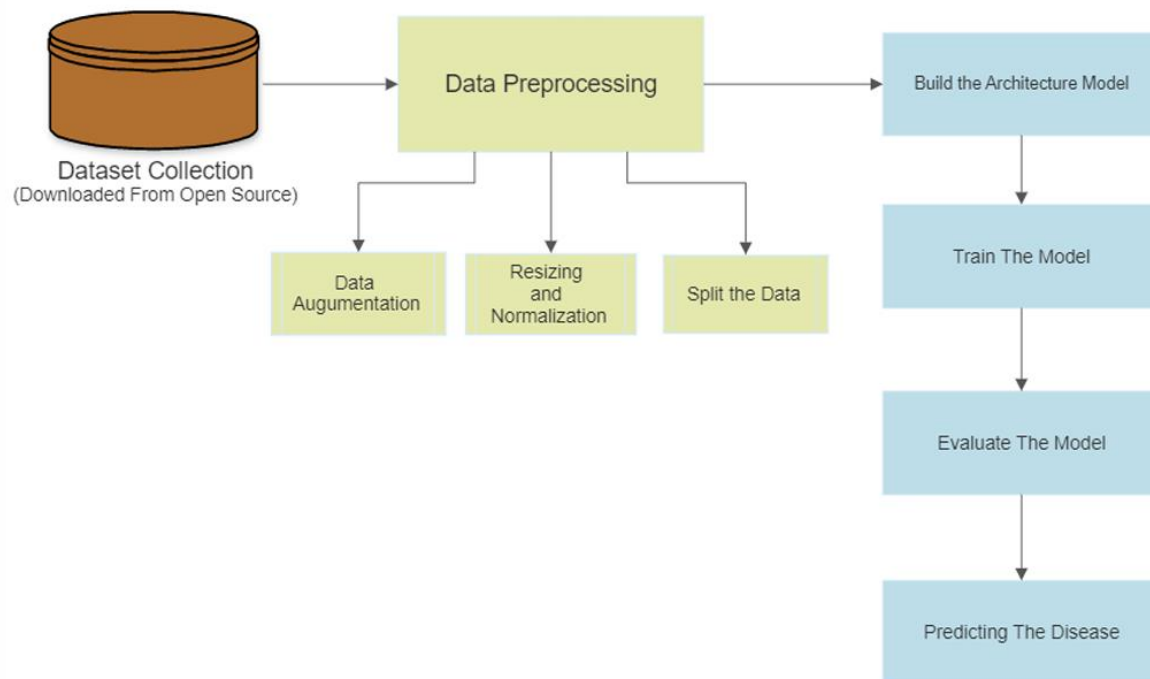
EXISTING WORK:

The Team members have implemented the tea leaves disease detection, using dataset from Kaggle, completed testing with new instances of tea leaves images. Implemented steps to improve accuracy, compared and drawn comparison about 4 different algorithms and Hence, recorded results with graphical representations.

PROPOSED WORK/SYSTEM:

The team proposed to predict the type of the disease with dataset from Kaggle, inspired the usage of pre-existing models from proper study of different research papers. Apply different preprocessing, feature engineering, machine learning and hyper parameter tuning algorithms.

SYSTEM DESIGN/PROPOSED MODEL ARCHITECTURE:



TECH STACK USED:

1. Python
2. Google Collab/ Visual Studio Code
3. Python libraries:
 - Numpy
 - Pandas
 - Matplotlib
 - Tensorflow
4. Machine Learning Algorithms:
 - CNN
 - Transfer Learning

WORKING MODULES:

- 1.Data Collection
2. Project Breakdown and Planning
3. Data Preprocessing
4. Model usage Planning and Research
5. Building models.
6. Model training
7. Hyper Parameter Tuning, Accuracy Maintenance
8. Testing with New Image instances and Analysis
- 9.Documentation

WORKING MODULES DESCRIPTION:

1.Data Collection: The team referred the dataset from Kaggle.

Link: [Identifying Disease in Tea leaves \(kaggle.com\)](https://www.kaggle.com/datasets/abhishek119/identifying-disease-in-tea-leaves)

2.Project Breakdown: The team used the agile project management method, split down the tasks among individuals. Prioritised tasks, transferred knowledge. Integrated code from the individuals, applied changes and implemented successfully.

3.Data Preprocessing:

As soon as the dataset was downloaded, we first imported libraries for data Preprocessing:

1. Rescaling of the images: reduction of pixels to appropriate range
2. Splitting of data: splitting in the ratio of 80: 20
3. Data augmentation:
 1. Rotation
 2. Cropping
 3. Zooming
 4. Flipping

4. Model Planning:

The team researched across 15 papers to understand the usage of the models in various use cases, shortlisted to 4 models : CNN, VGG-19, NasNet , InceptionV3.

5.Model Building: each of three members, build the model referencing the research paper. Try different types of parameters, layers, values and finalize before finding accuracy.

6.Model Training: here we execute the training of the model by splitting data in the ratio of 80:20, 80 for training, 20 for validation.

7.Fine tuning/Hyper parameter tuning: In this step we fine tune the layer parameters, transfer learning parameters for improvement of model accuracyHyperparameters include learning rate, batch size, number of layers, f ilter size, activation functions, and regularization approaches.

8.Testing and Analysis: in this step , we provide new instances of images to the model for predicting the diseases and record the results. Compare results, accuracy between different models. Calculate the confusion matrix and analyze the results

9. Documentation: Final step of the project is to document the results, code snippets before presentation

OUTPUT:

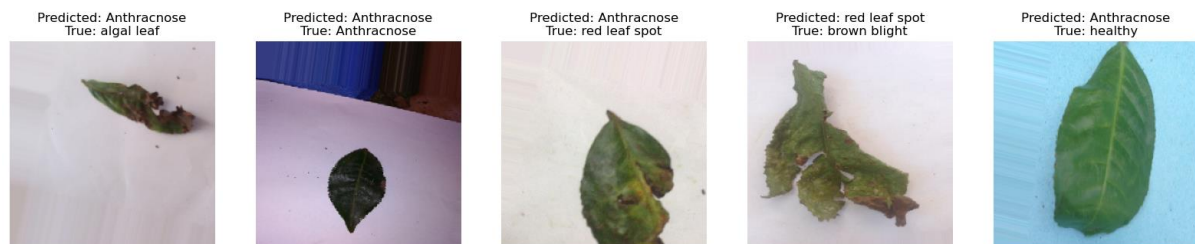
CNN



NASNETMOBILE



INCEPTIONV3



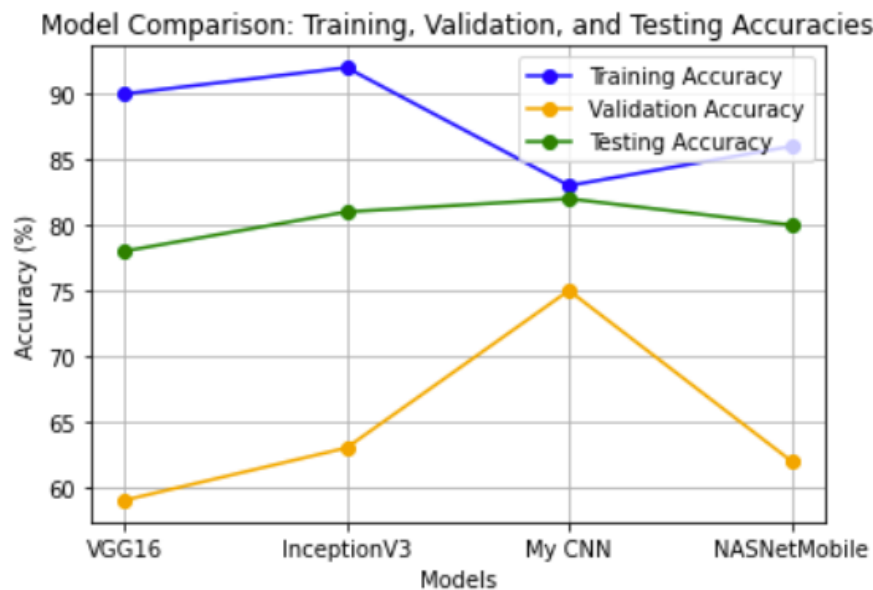
VGG16



CONCLUSION:

In a comparative evaluation of various pre-trained architectures alongside a CNN-based model, it was observed that the CNN-based model surpassed NasNetMobile, VGG-16, and Inception V3 in performance. The customized CNN-based model demonstrated superior performance and was also noted for its ease of training, attributed to its lower number of trainable parameters and reduced computing complexity compared to the other pre-trained architectures.

Due to its effectiveness and ease of training, the customized CNN-based model is deemed more suitable for tea leaf disease detection tasks. The proposed model achieved commendable results with an 83% training accuracy and an 82% testing accuracy, underscoring its efficacy in identifying tea leaf diseases.



FUTURE WORK:

Looking ahead, there are exciting possibilities for improving our customized CNN-based model even further. We could explore ways to make it even smarter and better at spotting tea leaf diseases. One idea is to try out different ways of tweaking the model to see if we can make it learn even better from the data it has. We might also want to consider using knowledge from other fields or fine-tuning the model specifically for tea leaf diseases.

Another cool idea is to team up our CNN-based model with other models to create a super team. By combining different approaches, we might be able to make our detection system even more accurate and reliable. Plus, we could look into adding special features to help our model spot any new or rare diseases it hasn't seen before.

To really put our model to the test, we could take it out of the lab and into real tea plantations. By seeing how well it performs in different environments and with different types of tea plants, we can make sure it's ready for the real world. And of course, we'll want to get feedback from the experts – the people who work with tea every day – to make sure our model is user-friendly and actually helpful on the ground.

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COMPLETE IMPLEMENTATION:

<https://github.com/VishaalD07/Tea-leaves-Disease/tree/main>